

Anmol Panda

Portfolio: anmolpanda.github.io

Github: github.com/anmolpanda

Email: anmolp@umich.edu

Ph: +1 734 510 0138b

SKILLS

- **Research Expertise::** Data Science, Causal Inference, Machine Learning, Natural Language Processing, Statistical Analysis, Survey Research, Social Media Data, Computational Social Science, Public Opinion Analysis, Race and Ethnic Politics, Information Communication Technology and Development (ICTD)
- **Languages/Frameworks:** Python, Stata, C, Java, JavaScript, SQL, Bash, Scikit-Learn, TensorFlow, Keras, Tomotopy, R
- **Teaching Experience:** Data Science, Data Mining, Deep Learning (at University of Michigan), Object Oriented Programming, Effective Public Speaking (at BITS Pilani)

EDUCATION

- **University of Michigan (Expected Graduation: May 2026)** Michigan, USA
Ph.D. - Information Science (GPA: 3.925) Aug 2020 - Present
Courses: Data Mining, Natural Language Processing, Categorical Data Analysis, Survey Methods, Multilevel Modelling, Public Opinion Seminar, American Political Development
- **Birla Institute of Technology and Science - Pilani** Goa, India
B.E. (Hons.) - Computer Science (GPA: 8.1) Aug 2012 - May 2016
Courses: Operating Systems, Data Structures, Design and Analysis Of Algorithms, Networking, Databases, Real Time Systems

WORK EXPERIENCE

- **Consultant for The World Bank** May 2022 - Aug 2022
Washington, D.C., USA
 - Consulted for the Water and Firms team at the World Bank
 - Analysed the Bank's gold-standard Enterprise Surveys of over 173K firms from over 150 countries using data science and statistical methods in Python and Stata
 - Studied factors that explain water shortages experienced by firms, geographical variation globally in duration and incidents of water shortages, and how demographics of firms impact their experiences
- **Research Fellow at Microsoft Research India** Jun 2018 - Aug 2020
Bangalore, India
 - Collaborated to lead the team that compiled a published the dataset of 33K Indian and 9.6K US politicians on Twitter using a combination of machine learning and human annotations
 - Analysed election campaigns, extreme speech, leader centrality, issue preferences of political parties, and Twitter trending topics
 - Published my findings at peer-reviewed (CSCW, Social Media and Society) conferences and mainstream media platforms
- **Graduate Student Instructor at the University of Michigan School of Information** Aug 2021 - Dec 2024
Ann Arbor, MI
 - Served as a GSI for Data Mining, Deep Learning, and Python Programming courses in in-person and online modes
 - Taught 2 tutorial sessions per week, each with 25-30 students, for Data Mining in Fall 2021 and Fall 2024
 - Conducted labs in Python with Jupyter Notebooks to teach matrix algebra, itemset mining, streaming mining, machine learning and natural language processing models, timeseries analysis, and deep learning models
 - Worked as a GSI for Python Programming, offering office hours, proctoring and grading services
- **Teaching Assistant at the ICPSR Summer Program** May 2023 - Jul 2025
Ann Arbor, MI
 - Worked for three summer terms as a teaching assistant for five different courses at the Inter-university Consortium for Political and Social Research (ICPSR) Summer Program
 - Assisted in the classroom, ran office hours, and supported students' projects and assignments for Data Science and Text Analysis (Summer 2023), Data Visualization (Summer 2024), Bayesian Models 2 (Summer 2024), Intermediate Mathematics (Summer 2025), and Network Science (Summer 2025).

- New Delhi, India
 - Developed compiler optimizations in software packet processing pipelines by leveraging memory level parallelism achieving up to 280 pc improvement in packet through put

- Prince George, BC, Canada
 - Tested and validated four GPS-free localization algorithms to build robots that can automatically seed farms

PROJECTS

- **Generative AI + Data Curation:** In this project funded by an IMLS grant, we explore how generative AI tools can be leveraged to create metadata for datasets published on large archives. Furthermore, we leverage variable descriptions to produce knowledge graphs that can aid in improved curation and search of data archives. We test this framework using GPT models hosted on Azure and datasets published on the ICPSR data archive. We aim to illustrate how generative AI can help mitigate the high cost and time for manual metadata generation and greatly enhance relevance and utility of search results in very large public archives.
- **Causal Effects of Digital Paywalls:** I analyzed the effect of introducing digital paywalls on newspaper content in the US. Combining data from 17 local newspapers and the US Census Bureau, we found that digital paywalls drive a decrease in local news across the US. This effect was greater in larger cities and cities that witnessed an influx of younger people during the study period. I used text data from the ProQuest Digital Archive, used BERT-based sentence transformer model to embed the data, and used K-Means Clustering to identify topical categories of news articles. We then used a staggered difference-in-difference model to assess the effect of paywalls on news content, and validated it using synthetic controls.
- **Social Media Use and Political Attitudes:** For my dissertation, I am studying the effect of social media platforms on attitudinal determinants of Trump support such as racial resentment, attitudes on immigration, sexism, and negative financial outcomes. I focus my attention on Asian American communities. I have developed a unique survey instrument that captures the effects of key variables on the racial attitudes of Asian Americans - a hitherto understudied group. The survey is currently in the field and is expected to be ready for analysis in Fall 2025.
- **Framing in Coordinated Campaigns on Social Media:** Using a dataset of tweets published by the UMass Amherst, I am analyzing the different frames used in networked publics on Twitter. We coded tweets for narrative frames like episodic and thematic, and theme-based frames like criticism of media elites, allegations of anti-Hindu bias, or allegations of pro-minority bias. We are using RoBERTa models to classify the tweets in our corpus. This project is a collaboration with Ceren Budak, Matt Bui, Zainab Akbar, and Julia Mendelsohn.
- **Social Media Archive:** I have built the archive of tweets of over 70K US and 60K Indian politicians and influencers hosted by the Social Media Archive (SOMAR) at ICPSR at UM. Presently, I am studying **how social media followers of political parties respond to changes in party coalitions at the state level** using a combination of community detection with **NetworkX** and topic modeling using multi-lingual **TensorFlow** language models. In the past, I have analyzed election campaign strategies, topical preferences in hashtag usage on Twitter, and misinformation spread by politicians.

MAJOR PUBLICATIONS

- **How digital paywalls shape news coverage (PNAS Nexus):** Paramveer Dhillon, [Anmol Panda](#), Libby Hemphill. Volume 4, Issue 1, January 2025, pgae511, <https://doi.org/10.1093/pnasnexus/pgae511>
- **Misinformation as a Window into Prejudice: COVID-19 and the Information Environment in India (CSCW'20):** Syeda Zainab Akbar, [Anmol Panda](#), Divyanshu Kukreti, Azhagu Meena, and Joyojeet Pal. 2021. Proc. ACM Hum.-Comput. Interact. 4, CSCW3, Article 249 (December 2020), 28 pages. DOI:<https://doi.org/10.1145/3432948>
- **Topical Focus of Political Campaigns and its Impact: Findings from Politicians' Hashtag Use during the 2019 Indian Elections (CSCW'20):** [Anmol Panda](#), Ramaravind Kommiya Mothilal, Monojit Choudhury, Kalika Bali, and Joyojeet Pal. 2020. Proc. ACM Hum.-Comput. Interact. 4, CSCW1, Article 053 (May 2020), 14 pages. DOI:<https://doi.org/10.1145/3392860>
- **NivaDuck - A Scalable Pipeline to Build a Database of Political Twitter Handles for India and the United States (Social Media and Society'20):** [Anmol Panda](#), A'ndre Gonawela, Sreangsu Acharyya, Dibyendu Mishra, Mugdha Mohapatra, Ramgopal Chandrasekaran, and Joyojeet Pal. 2020. In International Conference on Social Media and Society (SMSociety'20). Association for Computing Machinery, New York, NY, USA, 200–209. DOI:<https://doi.org/10.1145/3400806.3400830>
- **Affording Extremes: Incivility, Social Media and Democracy in the Indian Context. (ICTD'20):** [Anmol Panda](#), Sunandan Chakraborty, Noopur Raval, Han Zhang, Mugdha Mohapatra, Syeda Zainab Akbar, and Joyojeet Pal. 2020. In Proceedings of the 2020 International Conference on Information and Communication Technologies and Development (ICTD2020). Association for Computing Machinery, New York, NY, USA, Article 9, 1–12. DOI:<https://doi.org/10.1145/3392561.3394637>